From Robots to Cobots

⁇ 1. Difference between a Machine and a Robot

⁇ What is a Machine?

A **machine** is a device that helps humans do work easily by using mechanical power.

- It follows **fixed instructions**
- Does **not think or sense**
- Needs **human control**
- Performs **simple or repetitive tasks**

⁇ Examples:
Fan, Bicycle, Mixer, Washing Machine

⁇ What is a Robot?

A **robot** is a programmable machine that can **sense, process, and act automatically**.

- Works using **sensors + program + actuators**
- Can **take decisions based on input**
- Performs tasks with **speed, accuracy, and consistency**
- Can work **independently (autonomous)**

⁇ Examples:
Robot vacuum, Industrial robotic arm, Delivery robot

⁇ Core Difference (Concept-Based)

Feature Machine Robot

Intelligence No intelligence Has programmed intelligence Sensing

Cannot sense environment Uses sensors to sense Control Fully

human-controlled Automatic after programming Task Type Simple tasks

Complex and precise tasks

Adaptability Cannot adapt Can respond to changes

�� Important Concept: How a Robot Works

A robot must:

1. Sense

- Uses sensors (light, touch, distance)
- Collects information from surroundings

�� Example: A robot detects obstacles

2. Process

- Uses a controller (brain) to make decisions

- If a device **only follows fixed commands** → Machine
- If a device **senses + decides + responds** → Robot

�� Example:

- Traffic signal timer → Machine
- Self-driving car → Robot

�� 2. Cobots (Collaborative Robots)

�� Meaning of Cobots

Cobots (Collaborative Robots) are robots designed to **work with humans safely**.

�� Key idea: **"Human in the loop"**

- Humans and robots **interact and cooperate**
- Not fully independent like traditional robots

�� Features of Cobots

- Safe for **human interaction**
- Can work **side-by-side with humans**
- Easy to program and use
- Often part of a **multi-robot system**

�� Multi-robot system = many robots working together (and sometimes with humans)

�� Difference between Robot and Cobot

Robot	Cobot
Works alone	Works with humans
Kept in separate areas (safety)	Designed to be safe near humans
Less human interaction	High human interaction
Fully automated	Semi-automated (collaborative)

1. Human Interaction in Robotics

Human interaction plays an important role in modern robotic systems. Although robots can perform tasks automatically, humans provide guidance, supervision, and decision-making when required.

Importance of Human Interaction

- Ensures robots work according to ethical and safety guidelines.
- Helps train robots to perform complex tasks.
- Monitors robot performance and resource utilization.
- Provides feedback for continuous improvement.
- Handles situations that require human judgment and decision-making.

2. Multi-Robot System (MRS)

Definition

A **Multi-Robot System (MRS)** is a group of two or more robots that work together in the same environment and communicate through a network to accomplish a common goal.

Features of Multi-Robot Systems

- Multiple robots operate simultaneously.
- Robots are interconnected through a communication network.
- Information is shared among robots.
- A human operator can monitor and control the robots remotely.
- Tasks can be completed faster and more efficiently.

Working of a Multi-Robot System

1. Human operator sends commands through a network.
2. Robots receive instructions and perform assigned tasks.
3. Robots communicate with each other.
4. Robots send real-time data and status reports back to the operator.
5. The operator monitors and coordinates robot activities.

Example
Firefighting Robot Team

In a burning building:

- Multiple robots search for survivors.
- Robots use sensors to detect fire, smoke, and obstacles.
- Some robots extinguish fires.
- The firefighter monitors all robots remotely.
- The firefighter coordinates robot movement and receives updates.

Advantages of Multi-Robot Systems

- Faster task completion.
- Better coverage of large areas.
- Improved reliability.
- Increased efficiency.
- Reduced human risk in dangerous environments.

3. Robot

Definition

A **Robot** is a programmable machine that can:

- Sense its environment.
- Process information.
- Make decisions based on programming.
- Perform different tasks automatically.

Characteristics of Robots

- Programmable
- Autonomous or semi-autonomous
- Uses sensors and actuators
- Performs repetitive or dangerous tasks

Applications

- Manufacturing
- Healthcare
- Space exploration
- Agriculture

- Defense

4. Collaborative Robot (Cobot)

Definition

A **Cobot (Collaborative Robot)** is a robot specifically designed to work safely alongside humans in a shared workspace.

Key Features of Cobots

- Lightweight
- Flexible
- Easy to program
- Equipped with safety sensors
- Energy efficient
- Can detect human presence
- Safe for human interaction

Safety Mechanisms

Cobots are designed with:

- Collision detection systems
- Force-limiting technology
- Proximity sensors
- Emergency stop mechanisms

These features help prevent injuries when working near humans.

5. Why Were Cobots Developed?

The development of cobots was driven by:

Increasing Demand for Automation

Industries require automated systems to improve productivity.
Human-Robot Collaboration

Organizations need robots that can work directly with human

workers. **Cost Reduction**

Cobots are generally less expensive than traditional industrial robots.

Lightweight and Agile Design

Cobots can be easily moved and adapted to different tasks.

Improved Workplace Safety

They reduce risks while maintaining efficiency.

6. Difference Between Robots and Cobots

Feature	Robot	Cobot
Purpose	Automate dangerous, repetitive, or tedious tasks	Work collaboratively with humans
Design	Large, heavy, often enclosed	Smaller, lighter, open design
Speed	Very fast	Usually slower
Strength	Very strong	Less powerful
Safety	Usually separated from humans	Designed to work safely with humans
Cost	More expensive	Less expensive
Flexibility	Less flexible	Highly flexible and adaptable

Exam Point

All cobots are robots, but not all robots are cobots.

7. Machine, Robot and Cobot

Machine	Robot	Cobot

	tasks	humans
Performs a specific task		
	Can sense and respond to environment	
Cannot sense environment		Can sense and safely interact with humans
	Works collaboratively with	
Performs multiple programmed		
Usually inanimate	Intelligent and programmable	Intelligent, programmable, and collaborative

Examples

Type	Example
Machine	Mixer Grinder, Sewing Machine
Robot	Industrial Welding Robot
Cobot	Collaborative Assembly Robot

Multi-Robot System :A multi-robot system contains several robots connected together to complete a task.

Example

Firefighting robots working together during a fire rescue operation.

Human Interaction in Multi-Robot Systems

Human operators:

Give instructions to robots
Monitor robot activities

Receive feedback

Ensure safety and efficiency

Benefits of Human Interaction

Better decision-making

Ethical control

Improved efficiency

Easier monitoring

Continuous learning and feedback